

Question ID: c7789423

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$|x - 2| = 9$

What is one possible solution to the given equation?

Correct Answer: 11, -7

Rationale

The correct answer is 11 or −7. By the definition of absolute value, if $|x - 2| = 9$, then $x - 2 = 9$ or $x - 2 = -9$. Adding 2 to both sides of the equation $x - 2 = 9$ yields $x = 11$. Adding 2 to both sides of the equation $x - 2 = -9$ yields $x = -7$. Thus, the given equation, $|x - 2| = 9$, has two possible solutions, 11 and −7. Note that 11 and -7 are examples of ways to enter a correct answer.